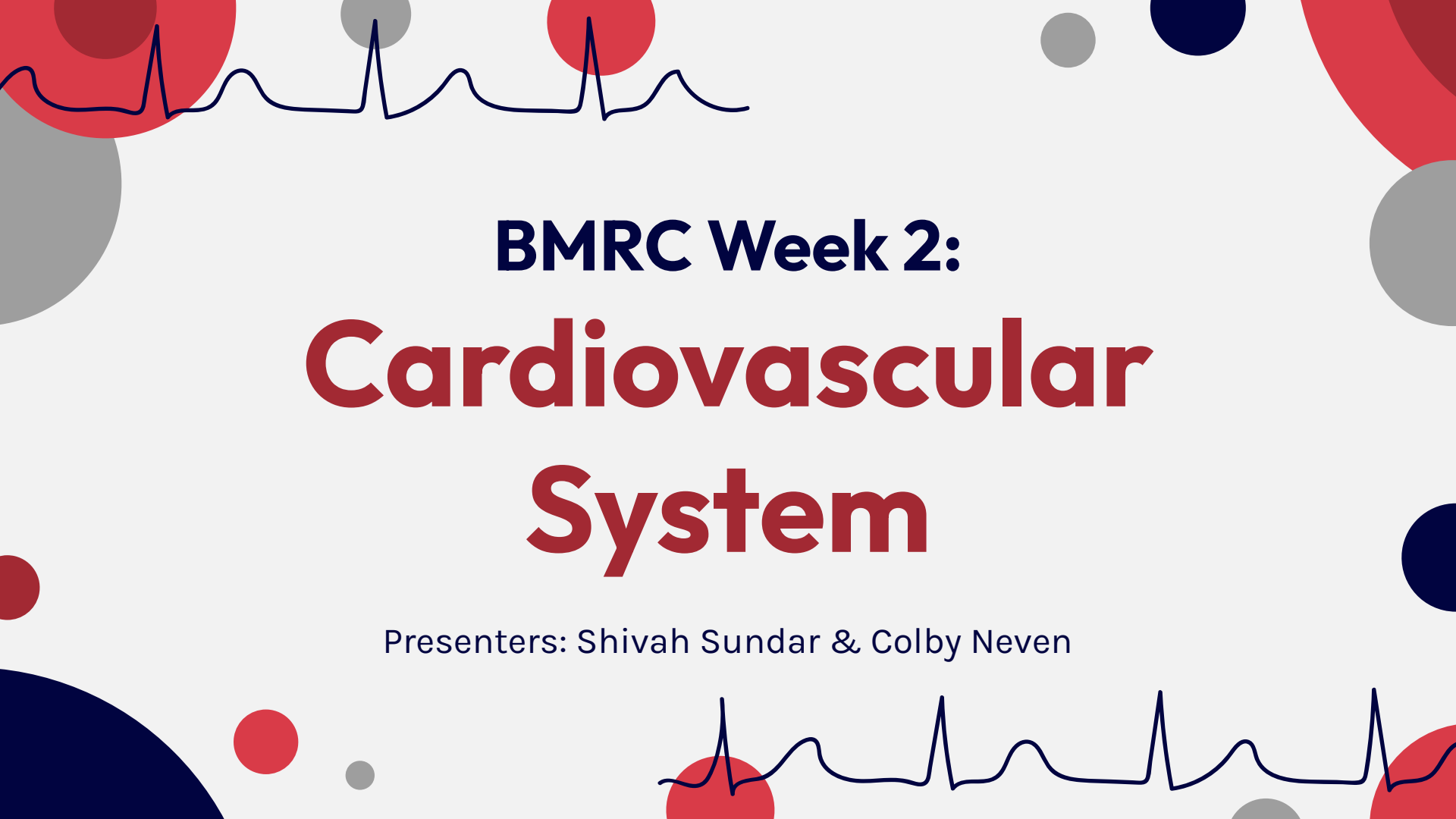




BMRC Week 2:

Cardiovascular

System

Presenters: Shivah Sundar & Colby Neven

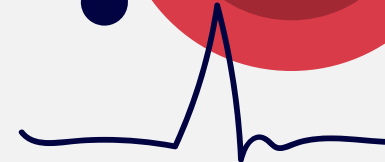
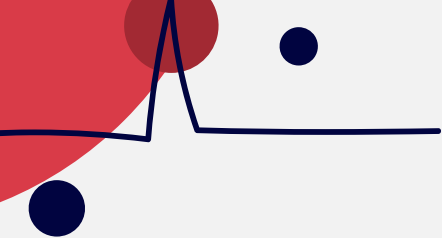


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01

CV Anatomy

Colby

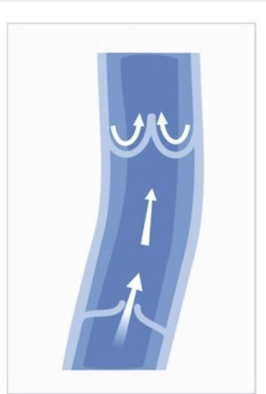
Parts of the CV System

Heart

The muscular organ that provides the force to pump blood, oxygen and nutrients through your body.

Veins

Vessels that carry oxygen poor blood from your body tissues back to your heart. They also have valves to prevent backflow.

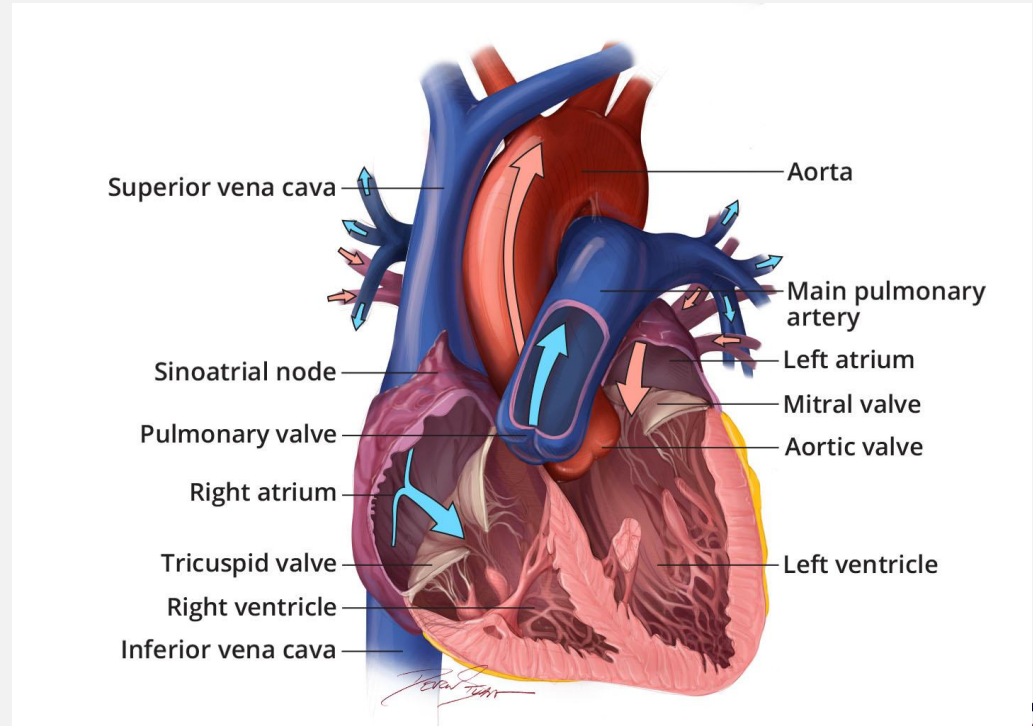
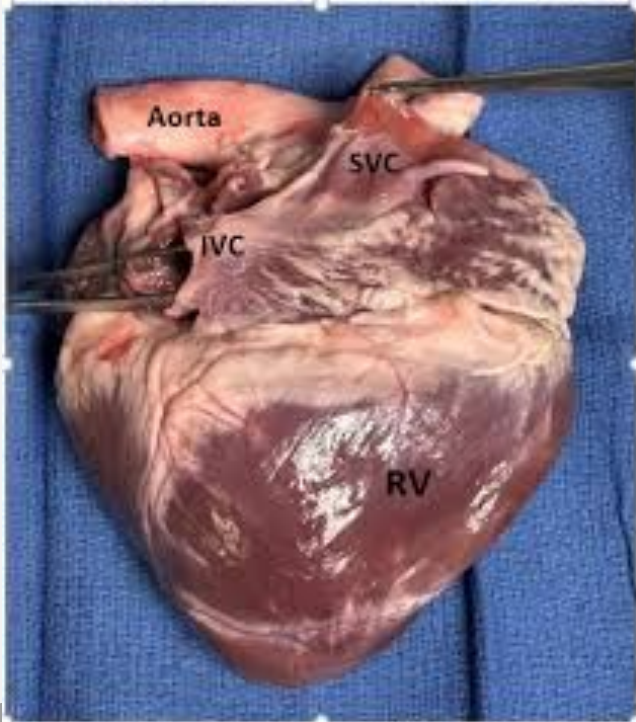


Healthy vein

Arteries

Arteries are muscular blood vessels that carry blood away from the heart to the rest of the body. They are thicker because they are under pressure from the heart pumping.

Anatomy of the Heart



Blood Vessels

Arteries: carries blood away from the heart

Arterioles: smaller arteries that carry oxygenated blood to the tissues of the body.

Capillaries: Small vessels that where gas and nutrient exchange occurs.

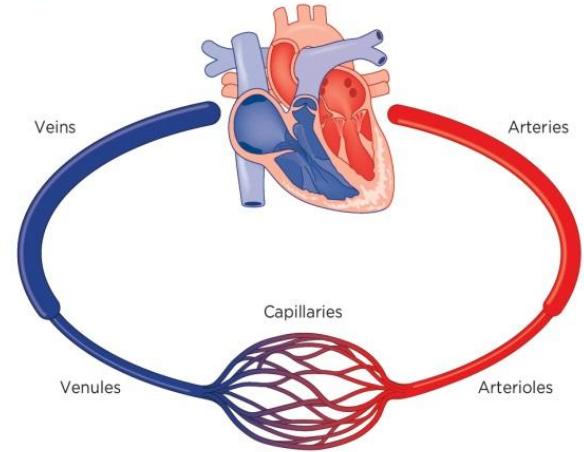
Venules: Collect blood from the capillaries to r

Veins: Return the blood back to the heart

The flow goes like so:

Heart > Arteries > Arterioles > Capillaries > Venules > Veins

Fig 1. The five types of blood vessels



Cardiovascular Circulation

The circulatory system is the movement of blood throughout the body.

The function:

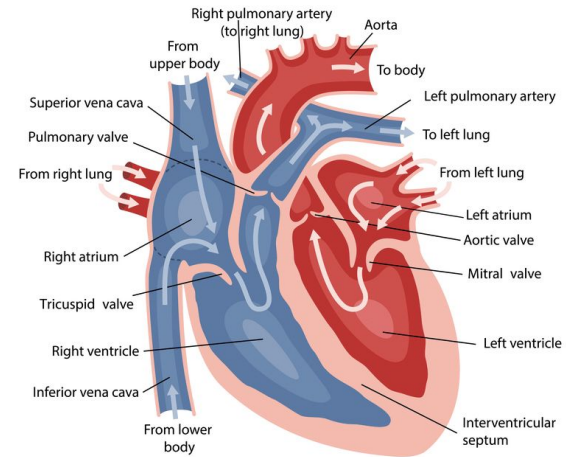
Pump oxygenated blood and deliver nutrients to the body (for uptake by capillaries systemically)

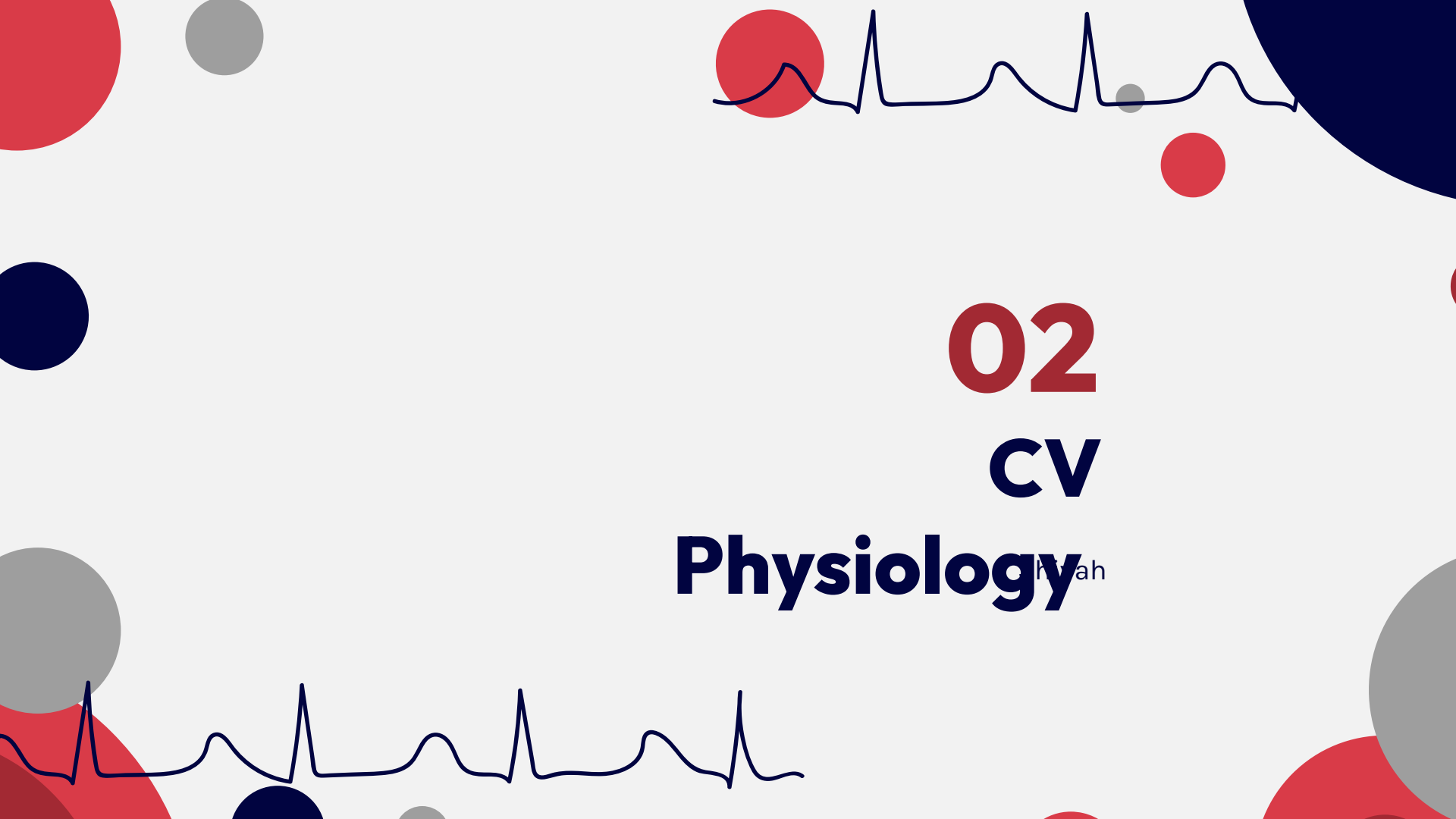
Remove deoxygenated blood and waste

The pathway:

(de O₂) inf/sup vena cava > right atrium > tricuspid valve > right ventricle > pulmonary valve > lungs > pulmonary vein > left ventricle > mitral valve > left atrium > aorta > systemic

Circulation of blood through the heart





02

CV

Physiology

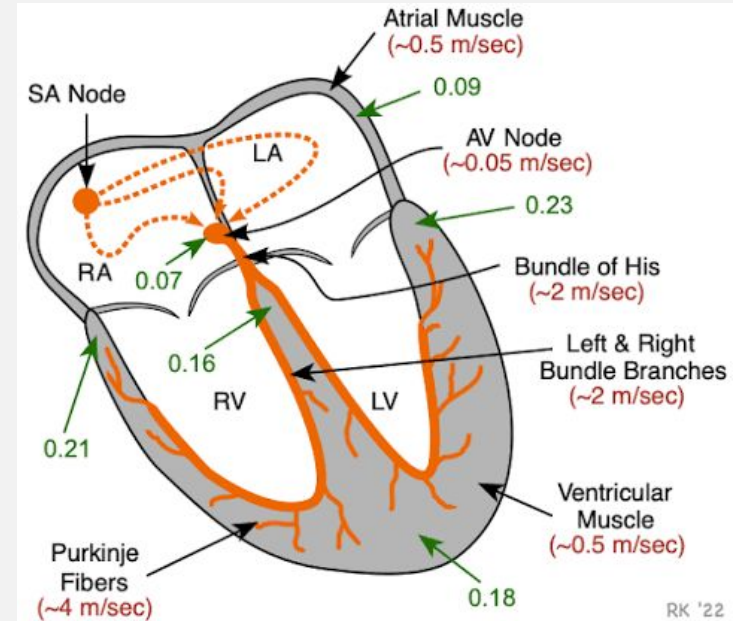
Physiology of the Heart

The Heart is an Electrical Organ

The heart operates independently from the Central Nervous System (CNS), but may be influenced by it

The heart operates thanks to an extensive electrical system originating from the Sinoatrial Node (or SA node)

The SA node acts as the heart's electrical pacemaker



Physiology of the Heart

A Free Quick Reference Guide From: **Nurse on Fire** Download for free at: NurseOnFire.com

Basic EKG/ECG Rhythms

Common & Fernald Rhythm Names	6 Second Rhythm Strip	Identifiers
V-Fib Ventricular Fibrillation		Irregular, No P Wave, No QRS
V-Tach Ventricular Tachycardia		Regular, No P Wave, Wide QRS
Torsade de Pointes Type Of Ventricular Tachycardia		Irregular, No P Wave, Wide QRS
*Synchronized Cardioversion possible for SVT if medication ineffective.		
SVT* Supraventricular Tachycardia		Regular, P Wave Hidden, Normal QRS
STEMI ST Elevation Myocardial Infarction		Reg or Irreg, P Waves, ST Elevated
A-Fib Atrial Fibrillation		Irregular, No P Wave, Normal QRS*
A-Flutter Atrial Flutter		Reg or Irreg, No P Wave, Normal QRS
PVC Premature Ventricular Contraction		Irregular, No P Wave, Wide QRS
Sinus Brady Sinus Bradycardia		Regular, P Wave, Normal QRS
Sinus Tach Sinus Tachycardia		Regular, P Wave, Normal QRS
NSR Normal Sinus Rhythm		Regular, P Wave, Normal QRS

*Medical Disclaimer Information Available At Nurse On Fire.com © Nurse On Fire 2016

The Heart is an Electrical Organ

Problems with the heart's electrical system lead to various disease conditions

Electrical rhythms are detected in the heart via the Electrocardiogram (or ECG/EKG for short)

Knowing the common normal & abnormal rhythms, and their causes is immensely valuable in EMS, even as an EMT

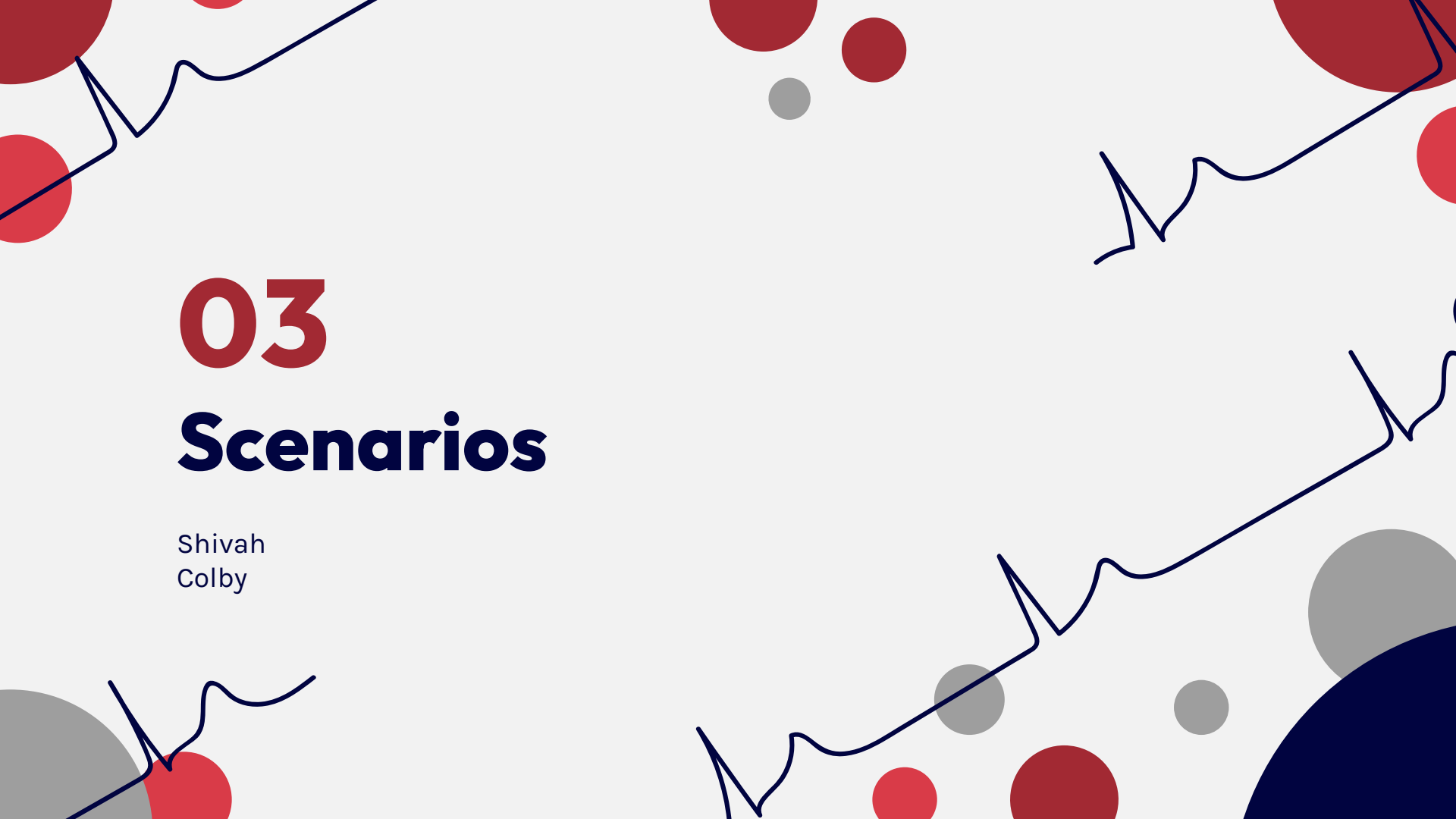
Physiology of the Heart

The heart receives & replenishes its own blood supply through its own set of blood vessels

These vessels are called Coronary Vessels

The heart receives its blood supply from the Ventricular End Diastolic Volume (EDV)

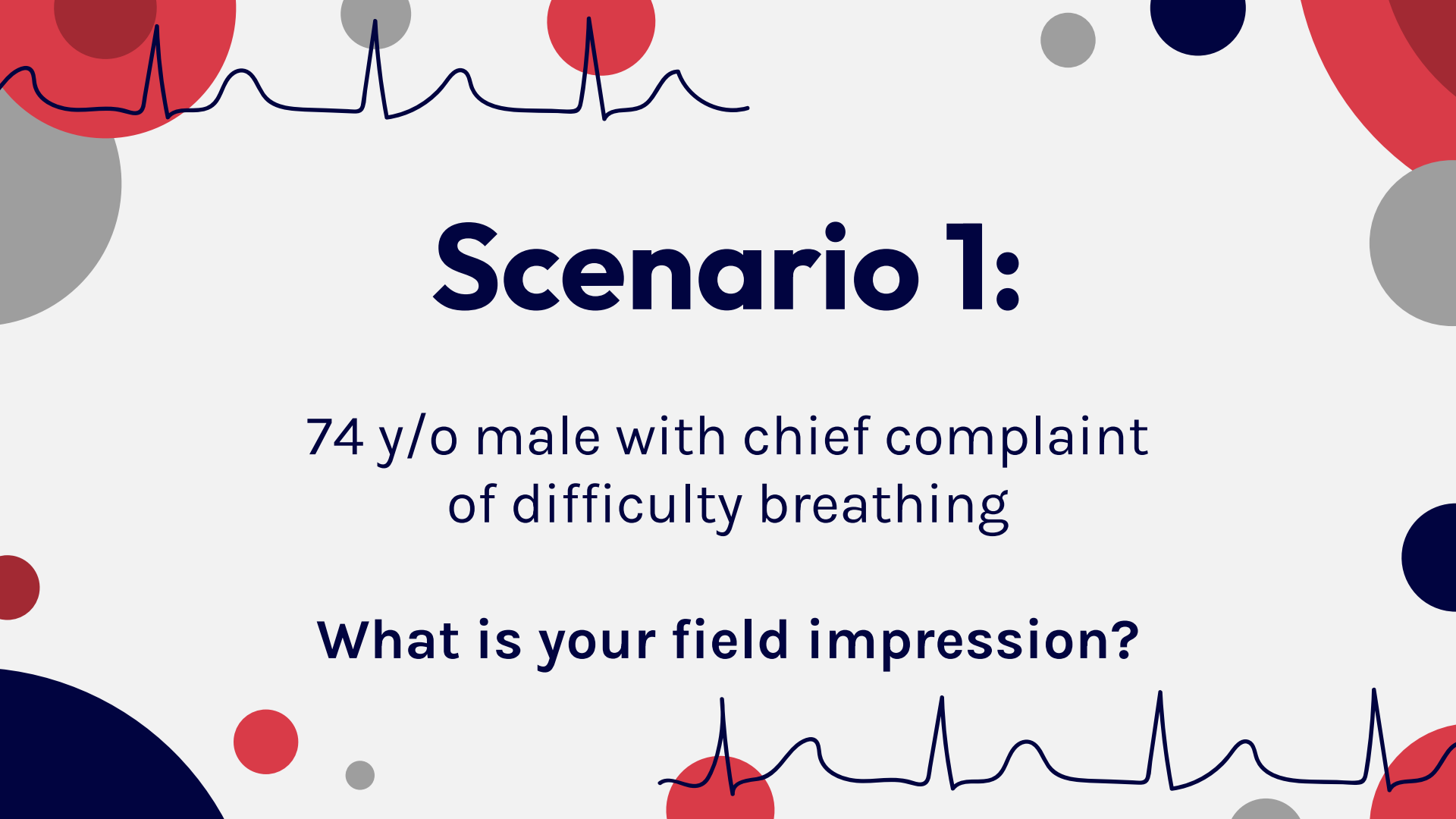
The freshest of the fresh



03

Scenarios

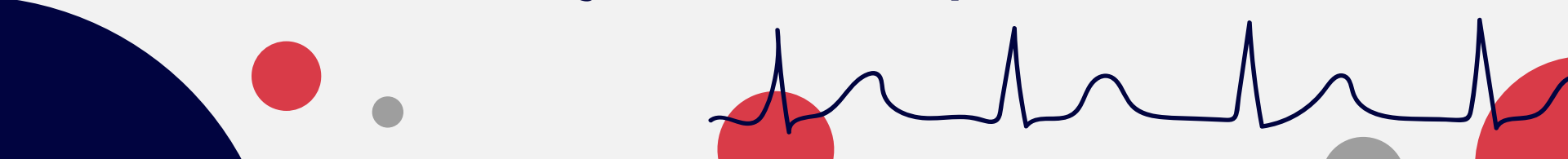
Shivah
Colby



Scenario 1:

74 y/o male with chief complaint
of difficulty breathing

What is your field impression?





Congestive Heart Failure

Congestive Heart Failure

Congestive Heart Failure

Congestive Heart Failure (CHF) is a secondary condition caused by inability of heart muscle to contract properly

Usually occurs secondary to an MI

2 types: Left Sided & Right Sided

Signs & Symptoms

Left Sided:

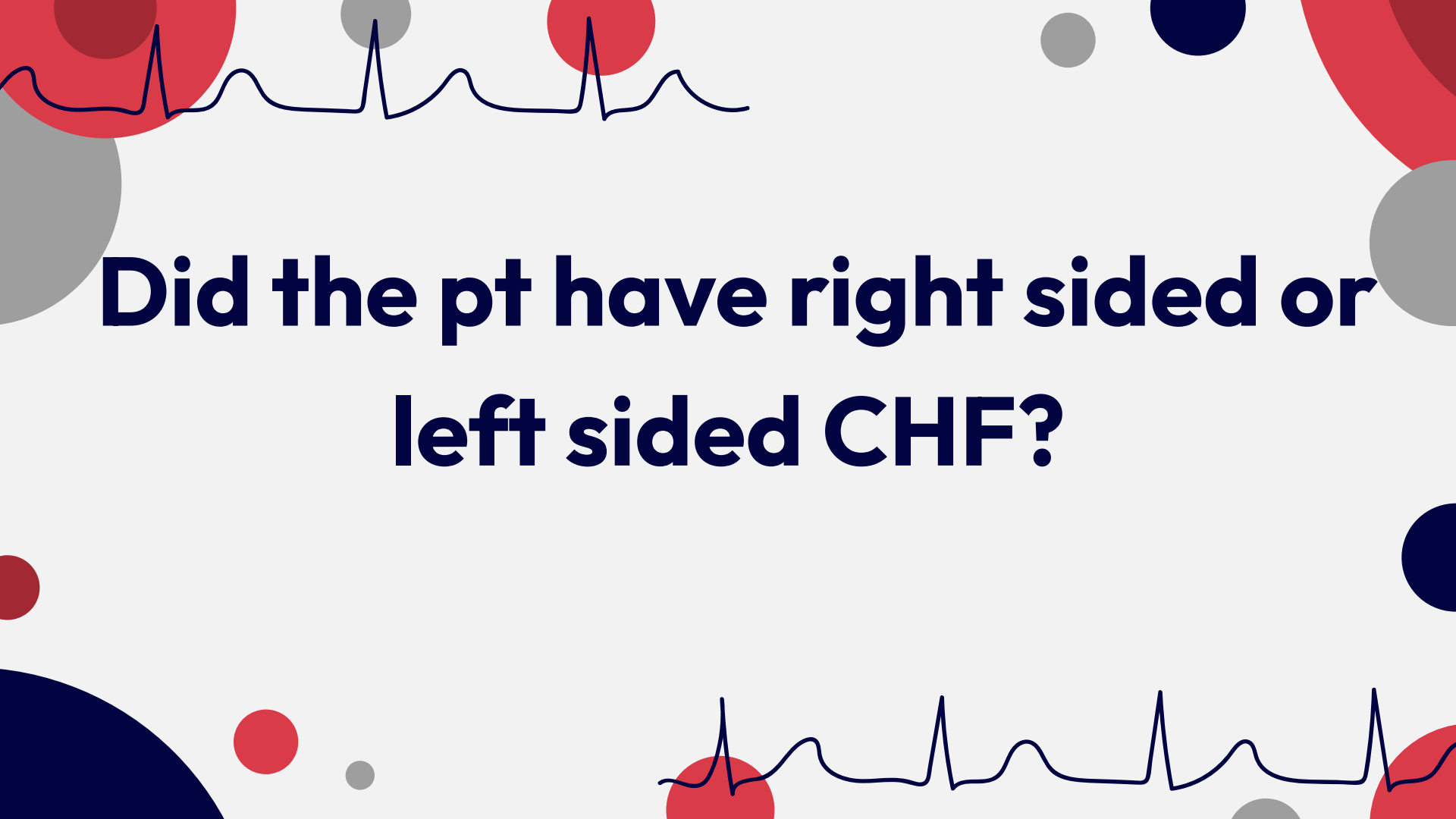
Signs: Low SpO₂, patient is tripodding, pedal edema in late stages

Symptoms: SOB, especially when lying down, coughing up pink, frothy sputum

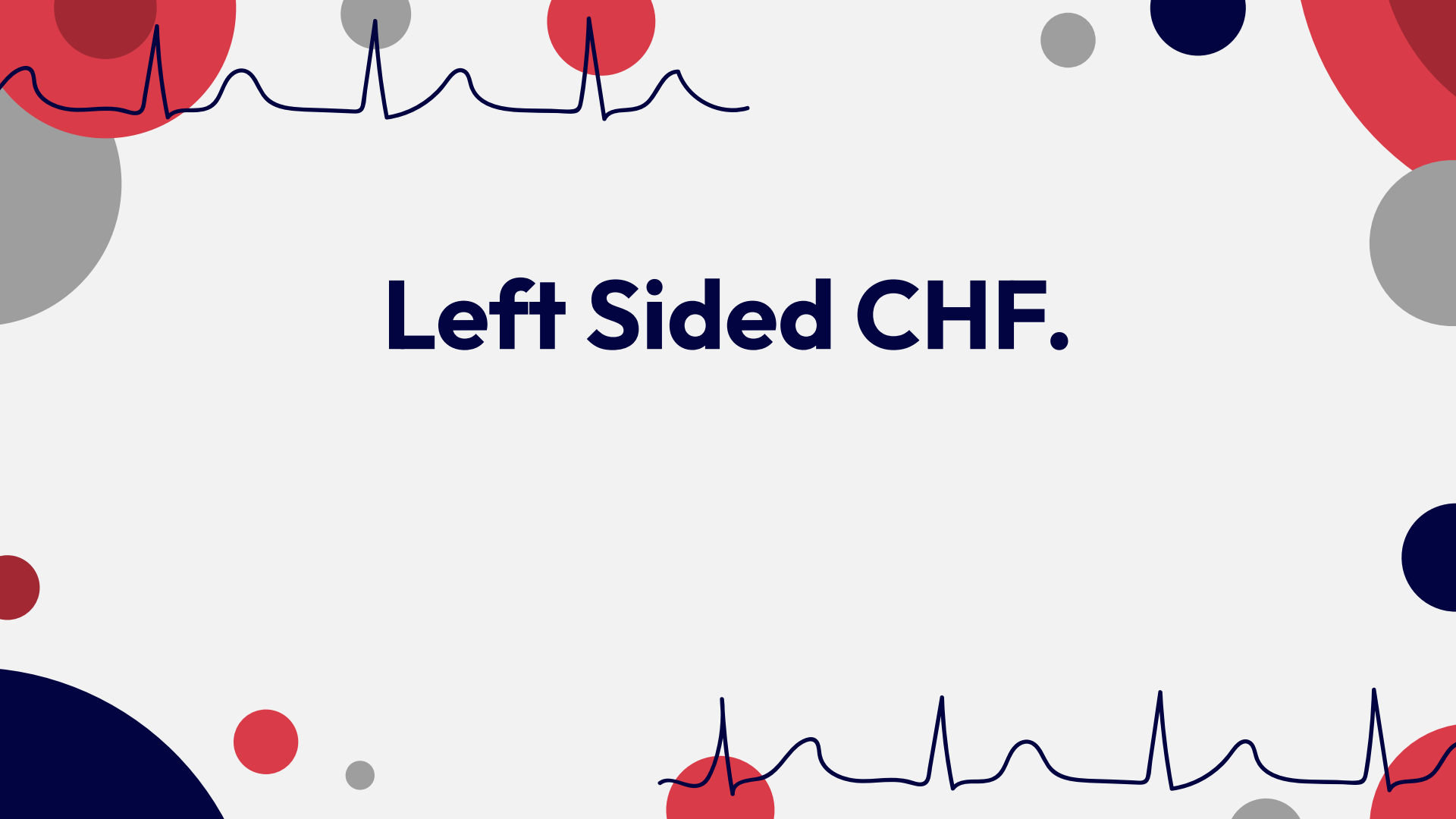
Right Sided:

Signs: Pedal edema, ascites (fluid in abdomen), JVD

Symptoms: Fatigue, SOB



**Did the pt have right sided or
left sided CHF?**



Left Sided CHF.

Congestive Heart Failure

Treatment

Provide Supplemental O₂, titrate to 94-99%

0.4 mg Nitroglycerin SL (if systolic >90 mmHg)

Provide positional comfort (especially for left sided CHF patients)

Rapid transport OR ALS intercept if possible

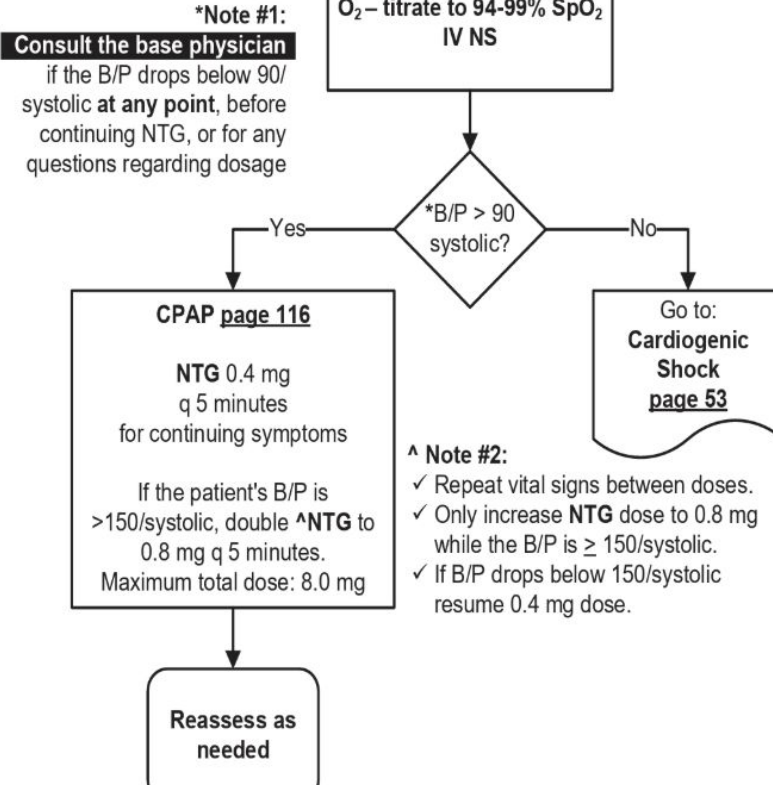
PULMONARY EDEMA / CHF

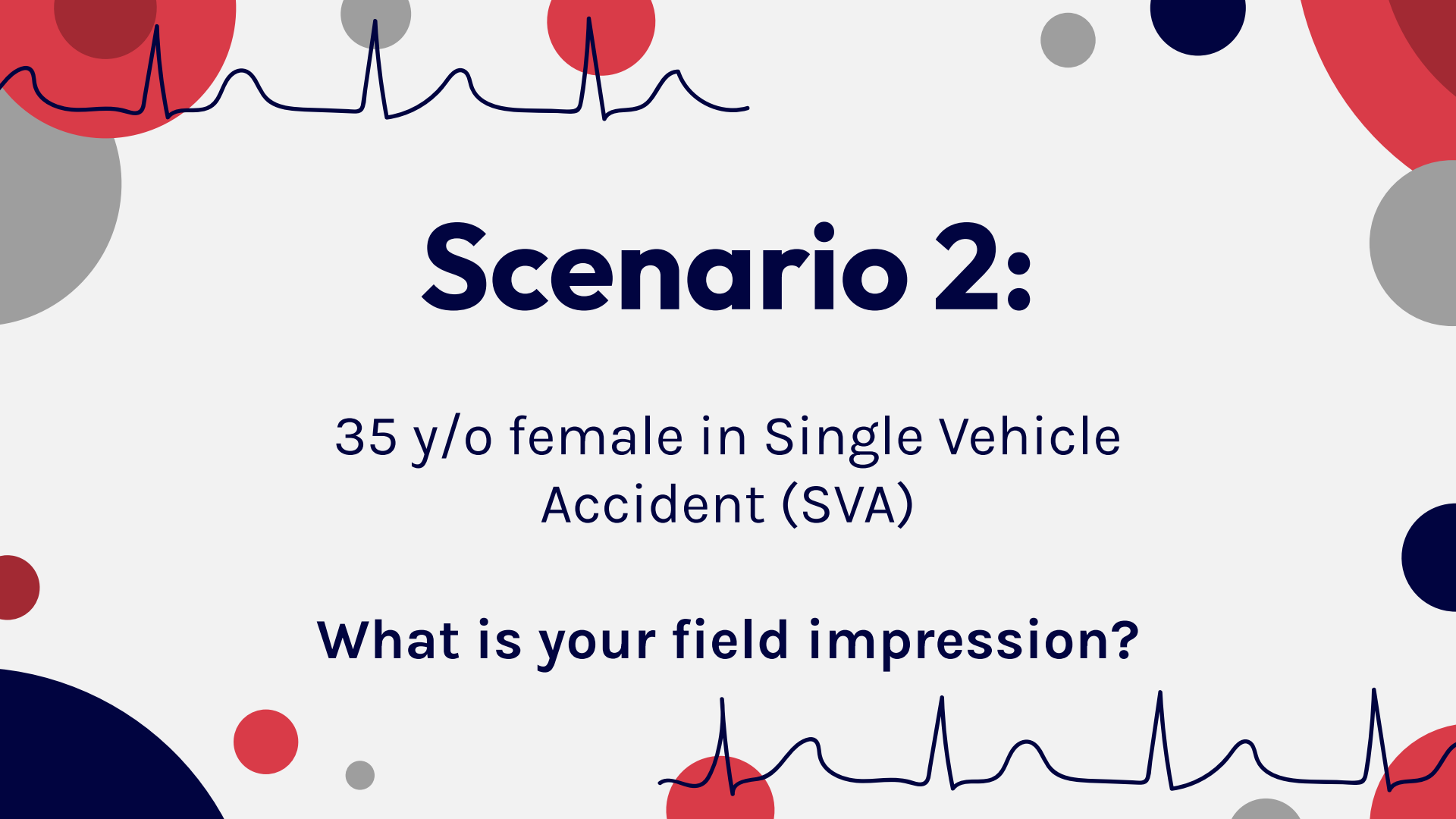
- **Routine Medical Care**

- **Consider ASA**, 162 – 324 mg po, for acute coronary syndrome patients

- Perform 12-Lead ECG, and transport to a STEMI Receiving Center if STEMI is identified. (See [page 120](#) - ECG 12-Lead) for STEMI Receiving Center information

- **Rapid transport** if on scene stabilization is unlikely

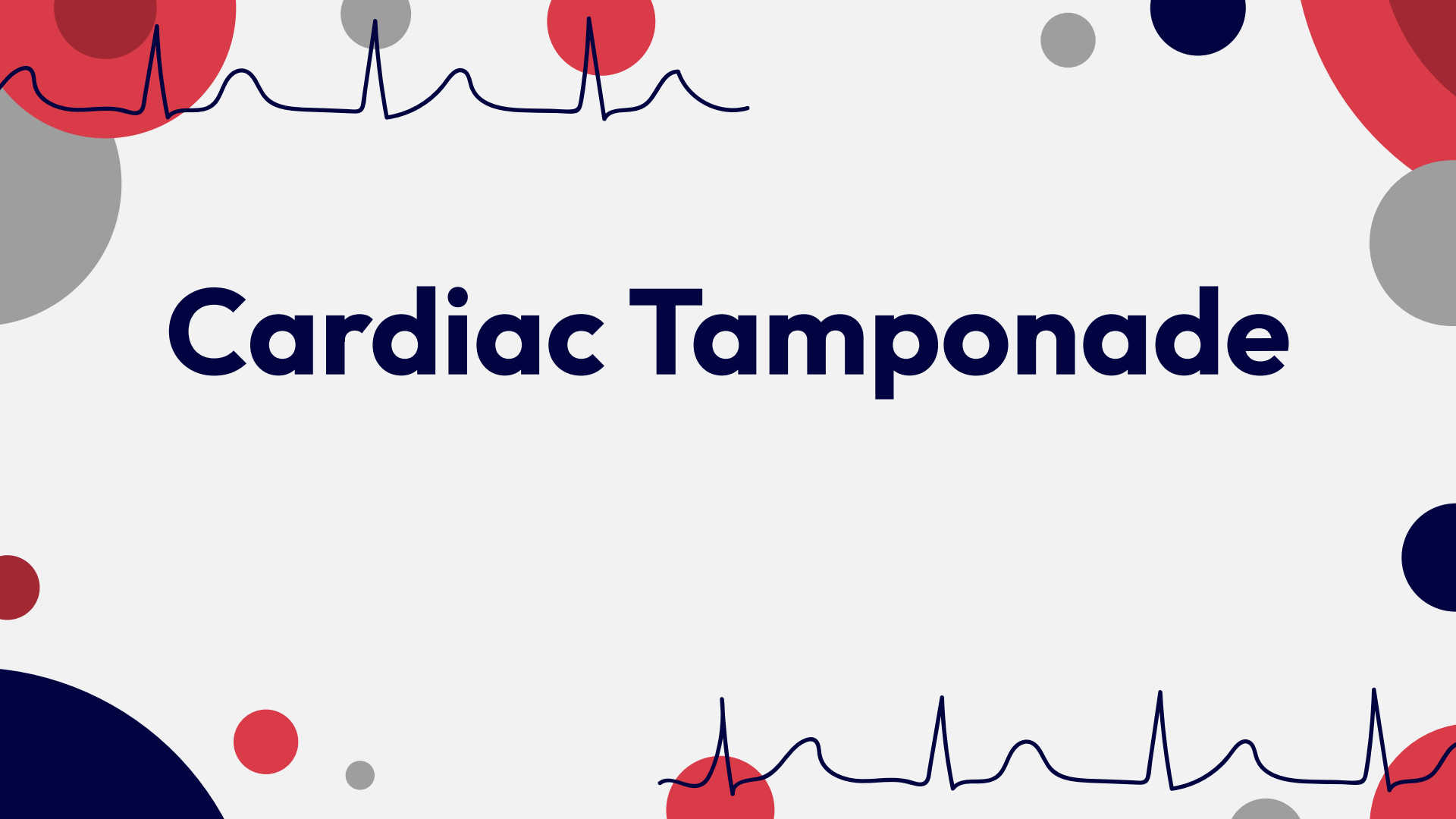




Scenario 2:

35 y/o female in Single Vehicle
Accident (SVA)

What is your field impression?



Cardiac Tamponade

Cardiac Tamponade

Cardiac Tamponade

A life-threatening condition caused by a blood or fluid build up in the pericardial sac which puts extreme pressure on the heart and prevents the heart from filling up with oxygen.

Can be caused by trauma, heart surgery complications, or cancer.

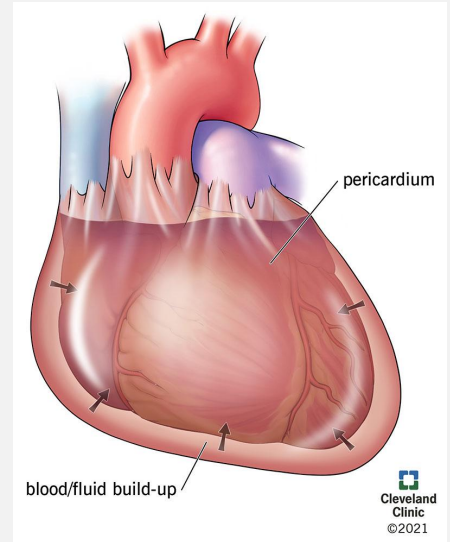
Known by Becks Triad: Hypotension, Distended neck veins (JVD), and muffled heart sounds.

Treatment

Requires an emergency pericardiocentesis which is a surgical procedure so rapid transport.

High flow oxygen to maximize oxygen delivery

Monitor airway & breathing



Bonus: Myocardial Infarction!

Myocardial Infarction (MI)

A myocardial infarction is the physiological term for a heart attack

An MI occurs due to a blockage of blood flow to heart muscle that results in damage (or ischemia) to heart muscle

A brief stoppage of blood flow to heart muscle that does not result in damage is known as Angina Pectoris

Signs & Symptoms

Signs: Pale skin, diaphoresis, patient is visibly anxious & clutching their chest in pain

Symptoms: Sense of impending doom, intense chest pain/pressure that radiates to the jaw & left arm, difficulty breathing

Myocardial Infarction

Treatment

- 1) Supplemental O₂ titrated to 94-99%
- 2) Aspirin 162-324 mg PO
- 3) Up to 3 doses 0.4 mg Nitroglycerin SL
 - Only if prescribed to patient
 - Only if patient has not taken ED drugs
 - Only if patient's BP is >90 mmHg systolic
 - Only administer additional dose
 - if patient's systolic BP drops <30 mmHg
- 4) Positional Comfort
- 5) Rapid Transport to a STEMI Center!

Note: MIs and Angina present very similarly so always treat with the urgency of an MI in mind. Time is tissue!

CHEST PAIN - SUSPECTED CARDIAC/STEMI

- Routine Medical Care

- Signs of Shock - 2 or more of the following:

- Pulse > 120/minute

- BP < 90/systolic

- Pale, cool and/or diaphoretic skin signs

- Altered Mental Status

- If cardiac chest pain is suspected and the patient is able to swallow, give **Aspirin 162 - 324 mg** po as soon as possible (tablet or chewable – not enteric coated)

- NTG may be prioritized as needed based on patient presentation

- Perform 12-Lead ECG, as appropriate, and transport to a STEMI Receiving Center if STEMI is identified. See [page 120](#) - ECG 12-Lead for ECG transmission and STEMI Receiving Center information

- **Note:** If the patient has taken **erectile dysfunction (ED) medication within the last 24 hours (Viagra/Levitra) or 36 hours (Cialis), withhold nitroglycerin**

Patients who have oxygen saturations of greater than 94% without signs or symptoms of hypoxia or impending airway compromise should not receive oxygen.

Monitor
Assess ABC's
O₂ – titrate to 94-99%
Aspirin 162-324 mg
IV/IO NS

12-lead EKG

* NTG 0.4 mg
up to 3 doses, q 3-5 minutes for
continuing pain/discomfort

If unresponsive to nitrates:
Pain Management
(see [page 42](#))

(^^see note)

^^ Note: If B/P drops below 90 systolic or drops > 30 mm/Hg from baseline at any point; or, **heart rate** is < 50 or > 120 bpm, **contact the base physician** before administering/continuing NTG and/or Pain Management



Condition #3

Arrhythmias

These are tricky, because you can only definitively treat 2 arrhythmias → Vfib & pulseless Vtach

But... it is good to know all of them, so you can help ALS

It is also good to know how to do a 12 lead EKG, and how to read a 12 lead EKG

Treatment

Will be covered when we talk about Cardiac Arrest (next slide)!

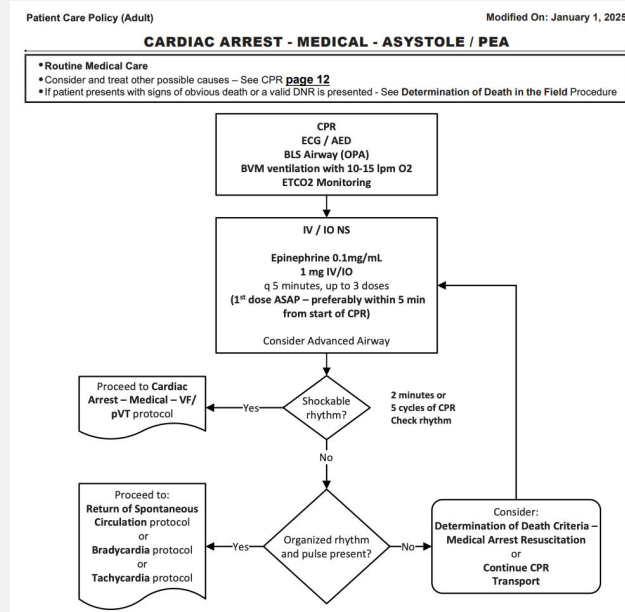
Condition #4

Cardiac Arrest

The main cause of cardiac arrest is a life-threatening abnormal heart rhythm which leads to a loss of all heart rhythm. Breathing stops, and the person becomes unconscious.

Is characterized by a loss of all heart activity and will be fatal if not treated within minutes.

Treatment



Condition #6

Hypertensive Crisis

A sudden rise in blood pressure to over 180/120 mmHg which puts a lot of strain on organs.

Internal organs are not used to this kind of mechanical and oxidative stress so organ failure is a possibility

Treatment

Give oxygen if spO_2 is under 94%

Requires immediate transportation

Condition #7

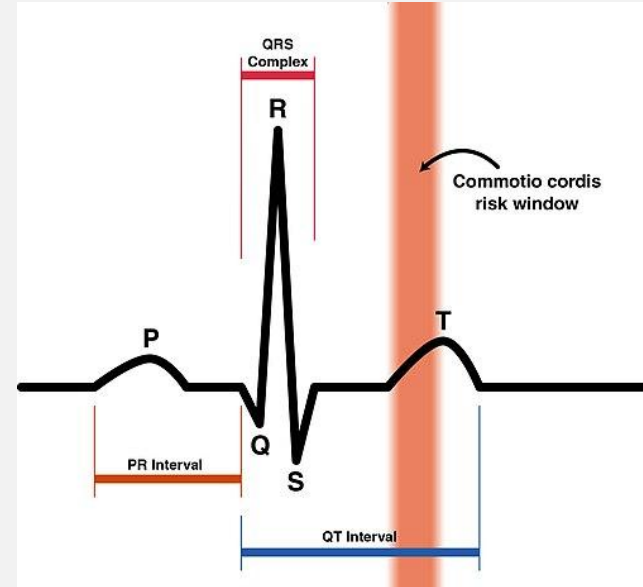
Commotio Cordis

Sudden cardiac arrest from a blunt trauma to the chest. Heart is structurally normal, and the heart enters ventricular fibrillation.

Usually caused during sports games like football, baseball, hockey and lacrosse.

Treatment

Immediate CPR
and AED
(shockable
rhythm)

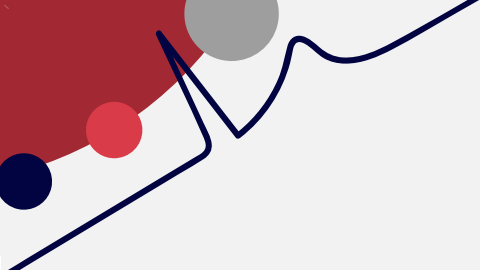




04

Diagnostic Process

Shivah



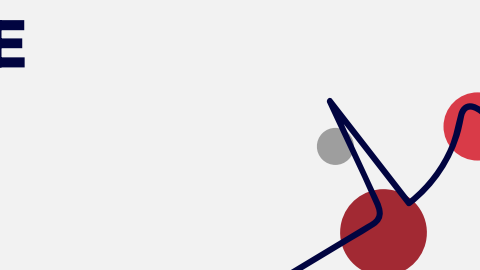
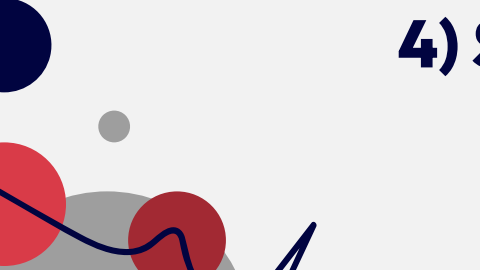
Diagnostic Process

1) Primary Assessment → ABCs

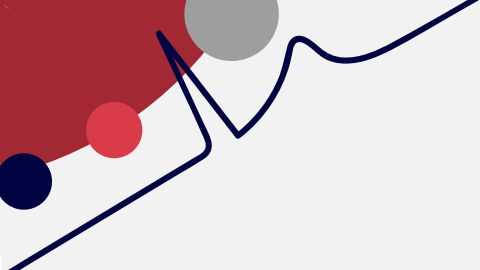
2) Transport Decision

**3) Medical History → CRUCIAL
SAMPLE, OPQRST**

4) Secondary Assessment → PASTE



5) Treatment/Differential



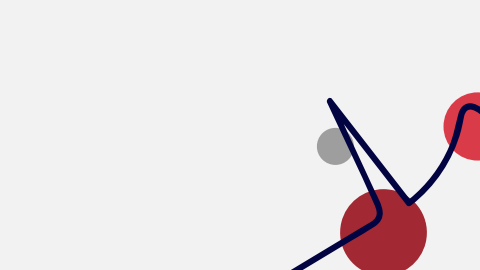
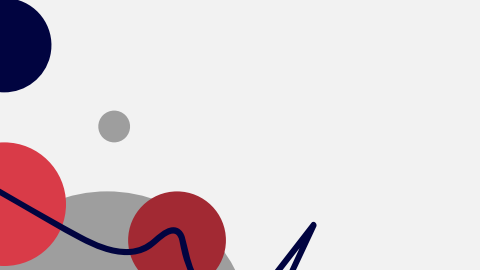
Diagnostic Process

**Keep in mind special cardiac assist
devices:**

VADs

ICDs

Pacemakers



Diagnostic Process

VENTRICULAR ASSIST DEVICES -VAD

OVERVIEW:

1. The VAD assists the native ventricle pumping action and provides the cardiac output needed to survive.
2. These devices are either pulsatile or continuous flow (non-pulsatile/pulseless). They are further divided into:
 - Left Ventricular Assist Devices (LVAD). The more common continuous flow pump located in the patient's thorax attached to the patients' left ventricle and aorta
 - Right Ventricular Assist Devices (RVAD),
 - Biventricular Assist Devices (BiVAD).

ASSESSMENT:

3. Assess for presence of a DNR, POLST or Advance Directive.
4. First ASSESS THE PATIENT, not the device.
 - The reason for the call may or may not be a problem with the VAD. VAD patients can and frequently do have other medical conditions.
 - Patients with a continuous flow VAD may have no discernible pulse or blood pressure.
 - ▶ Because there may be no palpable pulse, utilize other parameters for patient assessment (level of consciousness, skin signs, capillary refill, etc.)
 - ▶ Pulse oximetry may be unreliable.
 - Utilize the American Heart Association's C-A-B recommendations, with one addition:
 - ▶ C = Circulation / Connections and Function (device)
 - ▶ A = Airway
 - ▶ B = Breathing
 - ETCO₂ will read accurately and be useful in assessment.
5. Assess the device to see if it is working.
 - Information regarding the type of device, the implantation hospital, and/or the VAD Coordinator contact telephone number may be available by a tag on the device, on the refrigerator, or on a medical alert bracelet.
 - If a caregiver is present, utilize his/her knowledge. The patient and their caregiver are the experts on scene for all issues related to the VAD. Listen to their directions regarding VAD device management until you are able to contact the VAD Coordinator.
 - The VAD Coordinator can help you decide the best course of action regarding assessment of the equipment. NOTE: Only the base hospital is legally allowed to give orders regarding patient care.
 - If the patient has a continuous flow VAD (non-pulsatile / pulseless), auscultate the left upper quadrant of the patient's abdomen for the "hum" of the VAD, which can help direct the appropriate actions.
 - ▶ A pulsatile VAD will make an audible sound without auscultation.
 - ▶ Pulsatile VADs are usually older devices which pump blood via pulsatile mechanism, generating a peripheral pulse.
 - Determine if the device has power.
 - ▶ If the device has power it does not necessarily mean that it is working, so the previous step is very important.
 - ▶ If the device has power, you will see a green light on the HeartMate II, the most commonly implanted device
 - ▶ On the HeartWare device, the display will clearly tell you the Liters per Minute (LPM) of blood flow.
 - Check the VAD for secure connections and that the batteries are charged and functional.
6. Remain patient-centric. Check the VAD device as directed, but remain aware of how your patient is doing clinically. Deliver routine medical care as required.

Patient Care Policy (Adult)

Modified On: May 10, 2019

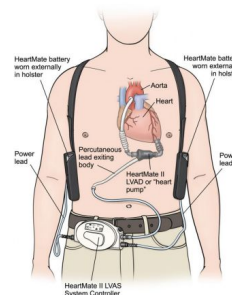
VENTRICULAR ASSIST DEVICES -VAD

- If the pump is pumping then the problem is usually with the patient, not the device.
- Do ABCs in conjunction with your VAD assessment.

TREATMENT/TRANSPORT:

7. If the patient's condition is related to their VAD, and it is safe and reasonable, it is preferred to transport the patient to their Bay Area VAD centers (Kaiser Santa Clara, Stanford, UCSF, and CPMC) unless the patient has any of the following conditions:
 - 7.1 MINOR medical or trauma patients with adequate perfusion: Follow appropriate protocol and transport to ANY basic ED or hospital of record.
 - 7.2 Suspected STROKE (STROKE ALERT) patient: Follow Acute Stroke policy and transport to closest Stroke Center.
 - 7.3 Suspected STEMI (STEMI ALERT) patient: Follow CP Suspect Cardiac/STEMI policy and transport to closest STEMI Center.
 - 7.4 Trauma patient (activation): Follow Trauma Care Policy and transport to closest Adult Trauma Center.
 - 7.5 Cardiac Arrest or critical / unstable patients (poor perfusion): Follow Shock or appropriate resuscitation policy and transport to closest STEMI / Cardiac Arrest Center.
 - 7.6 "Ring down" the receiving hospital early to help the facility prepare for this highly specialized patient.

VAD CENTER	24-HOUR HOTLINE
Stanford Hospital and Clinics	650-723-6661 (ask operator to page the VAD Coordinator- pager code #12502)
California Pacific Medical Center	415-600-1051
UC San Francisco	415-443-5823 (pager number)
Kaiser Santa Clara	408-851-3750
*Stanford Hospital and Clinics & Lucille Packard Children's Hospital at Stanford share the same VAD Coordinators	



05

Kahoot Time

and... ----->

JOIN US FOR A...

YOGURT PARK

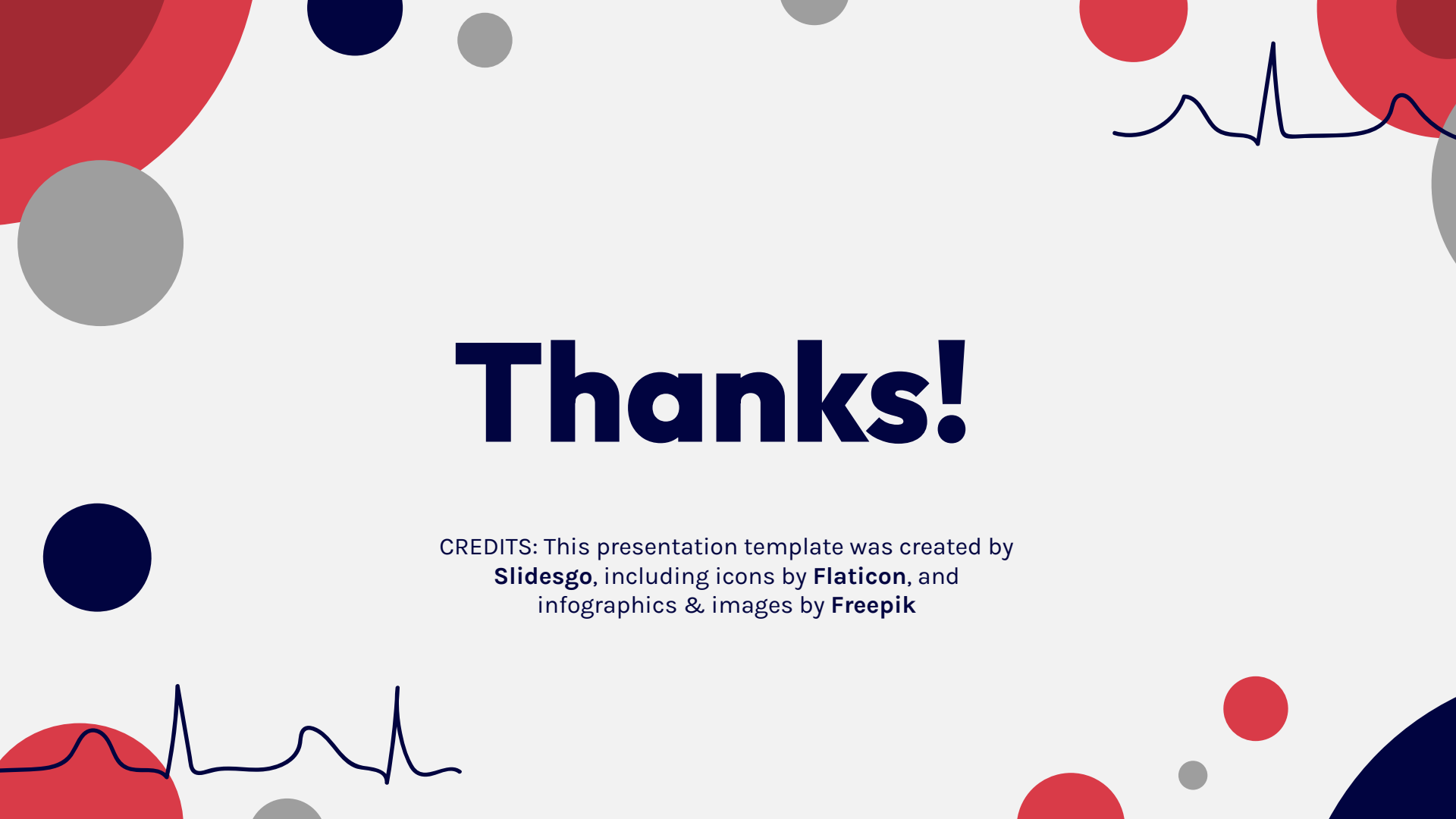


SOCIAL

AFTER MONDAY TRAINING

MONDAY SEP. 21ST 8 PM

WALK OVER FROM MOFFITT 145



Thanks!

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